

Wrong Shape Does Not Mean Biased

AP Statistics · Topic 5.5 (CED 3.2) · B: 2026-11-23 · E: 2026-12-21 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 3.C** — Describe probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.K** — Determine parameters of a sampling distribution for sample proportions. [Skill 3.B]
- **EK UNC-3.K.1** — For independent samples of a categorical variable from a population with proportion p , the sampling distribution of the sample proportion has mean $\mu = p$ and a standard deviation $\sigma = \sqrt{p(1-p)/n}$.
- **EK UNC-3.K.2** — If sampling without replacement, the standard deviation of the sample proportion is smaller than the formula above. If the sample size is less than 10% of the population size, the difference is negligible.
- **LO UNC-3.L** — Determine whether a sampling distribution for a sample proportion can be described as approximately normal. [Skill 3.C]
- **EK UNC-3.L.1** — For a categorical variable, the sampling distribution of the sample proportion will have an approximate normal distribution provided the sample size is large enough: $np \geq 10$ and $n(1-p) \geq 10$.
- **LO UNC-3.M** — Interpret probabilities and parameters for a sampling distribution for a sample proportion. [Skill 4.B]
- **EK UNC-3.M.1** — Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

Essential question: How do sampling, spread, and shape conditions limit what a surprising sample proportion can show?

DOK-3 skill: 4.B · **FRQ pattern:** failed-large-counts-normal-tail-vs-valid-binomial-and-bias

DOK rationale: Part (c) separates independence and shape conditions, compares two probability models, adjudicates distinct overclaims about evidence and bias, and bounds the strongest conclusion supported by the random sample.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask students to mark which sentence in each claim is about shape, center, or evidence.
- Begin the Topic 5.5 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and press the distinct roles of 10 percent and Large Counts.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose whose reasoning is closer and name one condition before the lesson.
- Complete the follow-along about the center, spread, shape, and probabilities for sample proportions.

- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which condition supports treating the 100 selections as approximately independent?
- Why does $np = 8$ address shape rather than the estimator's long-run center?
- What values of X belong to $\hat{p} \geq 0.14$?
- How do 0.0135 and about 0.028 change the strength, but not the direction, of the evidence?

Adult role

Solo teacher. Circulate 5 pairs. Require students to attach each condition to its job and to distinguish evidence against 0.08 from proof of a changed population rate.

[WARN] WATCH FOR

- Applying an $n > 30$ rule carried over from sample means.
- Calling a nonnormal sampling distribution biased.
- Calculating exactly 14 rather than 14 or more caption users.

SCORING PART (c) — E / P / I

Expected elements

- **distinguishes-two-condition-roles** (required) — Uses $100 < 500$ to support approximate independence and $np = 8 < 10$ to reject the normal approximation
- **uses-valid-tail-method** (required) — Uses a binomial calculation or a fully specified simulation to estimate the upper-tail probability near 0.028
- **bounds-evidence-conclusion** (required) — Calls the result unusual evidence under $p = 0.08$ without claiming it proves an increase
- **separates-normality-from-bias** (required) — Explains that failure of a shape condition does not shift the long-run center of the sample proportion and states the consequence of each overclaim

E	Correctly assigns the 10 percent and Large Counts conditions, obtains a valid tail estimate near 0.028, rejects both overclaims, and distinguishes unusual evidence, normal shape, and unbiased estimation with both reader consequences.
P	Gets the model choice and evidence conclusion but omits one condition role, gives an incomplete simulation, or explains only one student's overreach.
I	Uses $n > 30$ to justify normality for a proportion, declares the estimator biased because $np < 10$, or treats a small probability as proof the rate changed.

Common mistakes

- Checking only $n > 30$ instead of both expected counts
- Using $P(X = 14)$ rather than the upper-tail event $P(X \geq 14)$
- Saying the 10 percent condition determines normal shape rather than approximate independence and spread
- Equating a skewed sampling distribution with a biased estimator

[STAR] TODAY'S PROBLEM — annotated

Suppose a video platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE (5 min)

Whose reasoning is closer to correct, Jules's or Nora's? Name one condition or claim you would check.

Key: Not graded. Each student has one correct observation and one overreach.

Rules callout “CHECK INDEPENDENCE, THEN SHAPE (keep on the page)” is printed on the student sheet.

DOK 1 (a) Identify the population parameter, sample statistic, population size, and sample size. Verify the reported value of $\hat{p}$.

Key: The parameter is p , the proportion of all 5,000 active accounts that regularly use captions. The statistic is the sample proportion $\hat{p} = 14/100 = 0.14$. Thus $N = 5,000$ and $n = 100$.

DOK 2 (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.

Key: The sampling distribution has mean $\mu_{\hat{p}} = p = 0.08$ and standard deviation $\sigma_{\hat{p}} = \sqrt{0.08(0.92)/100} = 0.027129 \dots \approx 0.0271$. The 10 percent condition holds because $100 < 0.10(5,000) = 500$, so dependence from sampling without replacement is negligible. Large Counts does not fully hold: $np = 100(0.08) = 8 < 10$, while $n(1 - p) = 100(0.92) = 92 \geq 10$.

DOK 3 (c) Adjudicate both claims. Use a binomial calculation or valid simulation for $P(\hat{p} \geq 0.14)$ under $p = 0.08$. State the strongest conclusion warranted, explain why failed normality does not create bias, and give the consequence of each overclaim.

Key: Neither claim is fully correct. Nora rightly rejects the normal approximation because $np = 8 < 10$; $n > 30$ cannot replace both proportion counts. With $X \sim \text{Bin}(100, 0.08)$, $P(\hat{p} \geq 0.14) = P(X \geq 14) = 0.0282357 \approx 0.0282$; the without-replacement result is about 0.0269. This is unusual evidence worth investigating, not proof of an increase. The SRS and 10 percent condition support an unbiased $\hat{p}$; normality concerns shape, not center. Jules's 0.0135 makes the evidence look about twice as rare and “proves” erases sampling chance. Nora's claim could make a reader discard a legitimate random-sample estimate.

EXIT

One thing the video changed about my first take: _____